

ROLAND SINGH

Guwahati, Assam, India

✉ 24f2003132@ds.study.iitm.ac.in

🌐 linkedin.com/in/roland-singh-27679a283

🐙 github.com/RolandSingh

Education

Indian Institute of Technology, Madras

May 2024 – April 2028

Bachelors of Science in Data Science and Programming

Sikkim Manipal Institute of Technology, Majitar

July 2023 – June 2027

Bachelors of Technology in Computer Science and Engineering

Don Bosco Senior Secondary School, Guwahati

April 2021 – March 2023

AISSCE (Class 12)

Don Bosco Senior Secondary School, Guwahati

April 2009 – March 2021

AISSE (Class 10)

Experience

Summer Intern

May 2026 - Present

Indian Institute of Technology, Kharagpur

Onsite

- Selected for IIT Kharagpur's GRISHMA Research Fellowship Program, working under Prof. Subhadip Mukherjee on deep learning frameworks for imaging inverse problems.
- Implement TV and Fields-of-Experts regularizers, solving the lower-level reconstruction with Adam and computing upper-level gradients via the implicit function theorem.
- Design an input-convex neural network (ICNN) based regularizer, benchmarking it on MNIST classification, CT and MRI segmentation.

Research Assistant

Aug 2025 - Present

Sikkim Manipal Institute of Technology, Majitar

Onsite

ClearVision – A Physics-Informed Lightweight GAN for Real-Time Turbid Underwater Image Enhancement

- Conduct research under Dr. Ashis Pradhan, focusing on enhancing and restoring degraded underwater imagery using GANs to improve visual clarity and other computer vision-related tasks.
- Developed a lightweight (9.1M parameter) Turbid-GAN model utilizing a residual encoder-decoder with Triplet Attention for real-time underwater image enhancement.
- Achieved a 234-fold improvement in ORB feature matching on turbid imagery, enabling reliable visual odometry and navigation for AUVs in sediment-degraded environments.

DDL-Mamba – Geometric Residual Learning for 3D Medical Image Segmentation

- Designing a Delta-Mamba based framework under Dr. Palash Ghosal that incorporates Deep Delta Learning (DDL) modules into a U-Mamba architecture for accurate 3D medical image segmentation.
- Developed a Delta-Mamba framework by integrating Deep Delta Learning (DDL) modules into a U-Mamba architecture, optimizing 3D medical image segmentation on the ACDC and Synapse datasets.
- Stabilized deep encoder representations using L2-normalized, Softplus-constrained weights to enforce non-negative global slot amplitudes. Introduced a β -gate visualization method to enable direct, layer-wise interpretability of the network's segmentation outputs.

Summer Intern

May 2025 - July 2025

Indian Institute of Technology, Guwahati

Hybrid

- Worked under Prof. Natesan Srinivasan to address critical challenges in low-light computer vision, developing solutions applicable to surveillance systems.
- Developed deep learning models for facial emotion recognition and object detection in low-light conditions, achieving 81% validation accuracy for emotion classification and 0.75 mAP for object detection on challenging nighttime datasets.
- Implemented and evaluated multiple image enhancement techniques including CLAHE, Diffusion Models, and Zero-DCE++ to optimize preprocessing pipelines for downstream computer vision tasks.

- Fine-tuned YOLO v8 and pretrained CNNs (VGG16, ResNet50, EfficientNet) on custom datasets, integrating enhancement frameworks to improve detection performance in poorly illuminated environments.

Summer Intern

June 2025

National University of Singapore, Singapore

Onsite

- Conducted research in computer vision for autonomous systems under Dr. Amirhassan Monajemi, implementing state-of-the-art detection and tracking algorithms for critical infrastructure monitoring.
- Developed an autonomous computer vision system for underwater pipeline inspection using YOLOv8 and DeepSORT, enabling real-time detection and tracking of corrosion and structural damage on subsea infrastructure.
- Processed and annotated 1,775 frames from underwater video footage, implementing HSV segmentation and data augmentation techniques to train robust detection models under challenging aquatic conditions.

Publications

- **ClearVision: A Physics-Informed Lightweight GAN for Real-Time Enhancement of Turbid Underwater Imagery** (Accepted for Oral Presentation at *IEEE OCEANS 2026*, Sanya)
R. Singh, D. Das, A. Pradhan
- **AquaRead: A Two-Stage De-warp and Diffusion Prior for Distorted Text** (In preparation for submission to WACV, 2027)
R. Singh, D. Chatterjee, P. Ghosal, A. Pradhan
- **DDL-UMamba: Geometric Residual Learning for 3D Medical Image Segmentation** (Under Peer Review at *Pattern Recognition for Medical Imaging Journal*)
D. Das, R. Singh, P. Ghosal, A. Pradhan
- **Benchmarking GAN Architectures for Underwater Imaging: The Trade-off Between Model Compactness and Real-Time Latency** (Accepted for Presentation at IEEE GCON, 2026)
D. Das, R. Singh, P. Ghosal, A. Pradhan

Professional Affiliations

Student Member

Jan 2026 – Present

Institute of Electrical and Electronics Engineers (IEEE), Kolkata Chapter

- Member, IEEE Ocean Engineering Society.

Reviewer

IEEE Guwahati Subsection Conference (GCON), 2026

- Served as a reviewer for the peer-review process.

Co-Curricular Activities

Member

Dec 2024 - Present

IAESTE LC Manipal, Karnataka

Remote

- Actively involved in the International Association for the Exchange of Students for Technical Experience (IAESTE) Local Committee in Manipal, helping facilitate international internships for students.
- Received an offer to intern in Tunisia as part of the IAESTE Exchange Programme.

Projects

Women's Safety App

Aug 2024 - Nov 2025

- Developed a dual-component safety system combining a reactive Android app (SOS alerts, shake detection, emergency navigation) with a proactive ML engine that predicts area-wise safety scores using XGBoost and NLP-based crime data extraction.
- Built an NLP pipeline using Bidirectional LSTM to automatically extract and classify crime incidents from 5,000+ news articles, achieving 84.8% accuracy, and trained an XGBoost model with 99% accuracy to predict safety levels across geographic regions.
- Implemented "Safest Route" simulation algorithm that intelligently navigates users around high-risk zones using Google Maps API.

Skills

- **Programming Languages:** Python, C, C++, MATLAB
- **Libraries and Frameworks:** PyTorch, TensorFlow, Transformers, Keras, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Tools and Systems:** Git, Linux Shell (Bash), CUDA, HPC GPU Clusters
- **Development Environment:** Jupyter Notebook, Visual Studio Code, PyCharm
- **Databases:** MySQL, MongoDB
- **Miscellaneous:** LaTeX, Blender

Relevant Courses

- Machine Learning
- Deep Learning
- Computer Vision
- Digital Image Processing
- Soft Computing
- Data Structures in C++ and Python

Honors and Awards

- Awarded a 40% tuition fee waiver scholarship at Sikkim Manipal Institute of Technology based on JEE rank (2023).
- Black Belt in Taekwondo (ITF) and certified national instructor, with active participation in national and international workshops, tournaments, and events.